

Lab Title: Use case Model and project scheduling

1. Use Case –Diagram for your project

- a. List the actors, Use cases and relationships.

Example-

Primary actors

- **Customer** (includes Guest & Registered user)
- **Admin**
- **Inventory Manager**

Core use cases

- Browse/Search Products
- Register / Login
- Add to Cart
- View/Update Cart

Key relationships

- **Checkout** — *includes* → View/Update Cart, Apply Coupon, Pay for Order
- **Cancel Order, Return/Refund** — *extend* → Track Order
- **Assign Delivery** — *includes* → Send Notifications

- b. Draw the Use case diagram. Use tool (e.g. Draw.io or lucid chart).

- c. For each use case also describe the steps needed to perform before the use case.

Example

1) Register / Login

- *Basic*: Choose Register/Login → Enter email/password (or OAuth) → System verifies → User authenticated.
- *Alternate*: Wrong password → Show error + reset option. Duplicate email → Suggest Login.

2) Add to Cart

- *Basic*: Open product → Select quantity/variant → Add to Cart.
- *Alternate*: Stock insufficient → Show max allowed.

2. Describe the design methodology you are using for your project (SDLC, Agile, OOAD). Why is the selected methodology best suited for your project?

3. Schedule your project using- *Gantt Chart*

- To create a Gantt chart, start by breaking down your project into phases, then into activities, and finally into detailed tasks. Assign a duration to each task and identify its predecessor—the task that must be completed before it begins. Organize this information into a table with columns for activity, predecessor, and duration. Using this structured data, draw the Gantt chart to visualize the project timeline effectively. Demo table-

<i>Phases</i>	<i>Detailed Activity</i>	<i>Predecessor</i>	<i>Duration(weeks)</i>
<i>Analysis</i>	<i>A Define objectives</i>	-	<i>1</i>
	<i>B Conduct interview</i>	<i>A</i>	<i>2</i>
			
<i>Design</i>			

- Draw the Gantt chart